

# Week 2 Lab Assistant Prep

Number systems, truth tables, Raspberry Pi CLI, and Git workflow

Main focus: helping beginners understand binary/logic fundamentals and become confident using the Raspberry Pi terminal.

| Week 2 area       | What students do                                                       | Your priority                                                     |
|-------------------|------------------------------------------------------------------------|-------------------------------------------------------------------|
| Number systems    | Convert between decimal, binary, octal, and hexadecimal.               | Be ready to explain place value and grouping clearly.             |
| Binary arithmetic | Add/subtract binary, use two's complement, understand limits.          | Help with carry, borrow, bit length, and signed ranges.           |
| Truth tables      | Complete AND, OR, NOT, XOR, and compound expressions.                  | Clarify OR vs XOR and step-by-step evaluation.                    |
| Raspberry Pi CLI  | Navigate folders, manage files, use nano, inspect system/network info. | Debug folder mistakes, nano confusion, ping, and ifconfig issues. |
| Git               | Commit and push completed work to GitHub.                              | Check they are in the repo and actually push.                     |

# 1. What Week 2 is About

Week 2 introduces the foundations of how computers represent and process information, then moves into practical Raspberry Pi command-line work.

- Decimal, binary, octal, and hexadecimal number systems.
- Binary addition, subtraction, two's complement, fixed-point binary, and 8-bit limits.
- Truth tables and Boolean expressions.
- Basic Raspberry Pi terminal commands and Git submission workflow.

## Main lab assistant focus

Keep explanations simple. Most students will struggle either with base conversion logic or getting lost in the terminal.

# 2. Relevant Repo Files

```
week02/01_number_systems/01_number_conversion.md
week02/01_number_systems/02_binary_addition.md
week02/01_number_systems/03_twos_complement.md
week02/01_number_systems/04_fixed_point.md
week02/01_number_systems/05_limits.md
week02/01_number_systems/README.md
```

```
week02/02_truth_tables/01_truth_tables.md
```

```
week02/03_CLI/01_simple_terminal.md
week02/03_CLI/02_managing_files.md
week02/03_CLI/03_nano_editor.md
week02/03_CLI/04_basic_info.md
week02/03_CLI/05_network_info.md
week02/03_CLI/README.md
```

# 3. What Students Are Expected to Do

## Number systems

- Complete conversions between decimal, binary, octal, and hexadecimal.
- Practise binary addition and subtraction.
- Find two's complement by flipping bits and adding 1.
- Represent decimal fractions using fixed-point binary.
- Understand unsigned and signed 8-bit ranges.

## Truth tables

- Complete AND, OR, NOT, XOR, and a compound expression table.
- Match truth tables to logic expressions.
- Understand symbols such as x for AND, + for OR, and overline/NOT.

## Raspberry Pi CLI

- Use terminal commands for navigation, file management, nano editing, system info, and network info.
- Commit and push all completed files/screenshots to GitHub.

## 4. What You Should Personally Understand

| System      | Base | Digits   | Beginner explanation                                        |
|-------------|------|----------|-------------------------------------------------------------|
| Binary      | 2    | 0-1      | Used by computers because circuits represent two states.    |
| Octal       | 8    | 0-7      | Groups binary digits in threes.                             |
| Decimal     | 10   | 0-9      | The normal number system students already know.             |
| Hexadecimal | 16   | 0-9, A-F | Shorter way to write binary; groups binary digits in fours. |

Hex values to know:

A = 10 B = 11 C = 12 D = 13 E = 14 F = 15

### Simple explanation

Binary is base 2, so each place value is a power of 2. Decimal is base 10, so each place value is a power of 10. Hexadecimal is base 16, so each place value is a power of 16.

## 5. Expected Number Conversion Answers

| Task            | Answer                                              |
|-----------------|-----------------------------------------------------|
| 156 decimal     | Binary: 10011100   Octal: 234   Hexadecimal: 9C     |
| 101101 binary   | Decimal: 45   Octal: 55   Hexadecimal: 2D           |
| 745 octal       | Decimal: 485   Binary: 111100101   Hexadecimal: 1E5 |
| 3F9 hexadecimal | Decimal: 1017   Binary: 1111111001   Octal: 1771    |

### Methods to explain

- Binary to decimal: multiply each 1 by its place value, for example  $101101 = 32 + 8 + 4 + 1 = 45$ .
- Binary to octal: group bits in threes from the right, for example  $101101 \rightarrow 101\ 101 \rightarrow 55$  octal.
- Binary to hex: group bits in fours from the right, for example  $101101 \rightarrow 0010\ 1101 \rightarrow 2D$  hex.
- Decimal to binary: use powers of 2 or repeated division by 2.

## 6. Binary Addition and Subtraction

| Binary addition rule | Meaning   |
|----------------------|-----------|
| $0 + 0 = 0$          | No carry. |
| $0 + 1 = 1$          | No carry. |
| $1 + 0 = 1$          | No carry. |

| Binary addition rule | Meaning           |
|----------------------|-------------------|
| $1 + 1 = 10$         | Write 0, carry 1. |
| $1 + 1 + 1 = 11$     | Write 1, carry 1. |

| Task              | Expected answer                       |
|-------------------|---------------------------------------|
| $1011 + 1101$     | 11000                                 |
| $100110 + 101011$ | 1010001                               |
| $11101 + 10111$   | 110100                                |
| $1101 - 1010$     | 0011 (or 11 without leading zeros)    |
| $10110 - 1001$    | 01101 (or 1101 without leading zeros) |
| $10001 - 1110$    | 00011 (or 11 without leading zeros)   |

### Beginner explanation

Binary subtraction works like decimal subtraction, but you borrow 2 instead of 10. Binary addition carries when the result reaches 2.

## 7. Two's Complement

- Step 1: Flip every bit.
- Step 2: Add 1.
- Explain that two's complement is a common way computers represent negative numbers.

| Original | Flip bits | Add 1  | Two's complement |
|----------|-----------|--------|------------------|
| 101010   | 010101    | 010110 | 010110           |
| 110011   | 001100    | 001101 | 001101           |
| 1001     | 0110      | 0111   | 0111             |

### Common question

Why add 1? Flipping gives one's complement. Adding 1 completes the conversion to two's complement, which works well for binary arithmetic.

## 8. Fixed-Point Representation

Fixed-point binary means some bits are reserved for the whole-number part and some bits are reserved for the fractional part.

Fractional place values:

$$1/2 = 0.5$$

$$1/4 = 0.25$$

$$1/8 = 0.125$$

$$1/16 = 0.0625$$

$$1/32 = 0.03125$$

| Task                                            | Working                      | Expected answer |
|-------------------------------------------------|------------------------------|-----------------|
| 12.75 with 4 integer bits and 4 fractional bits | $12 = 1100$ , $0.75 = .1100$ | 1100.1100       |
| 5.125 with 3 integer bits and 5 fractional bits | $5 = 101$ , $0.125 = .00100$ | 101.00100       |
| 7.5 with 4 integer bits and 4 fractional bits   | $7 = 0111$ , $0.5 = .1000$   | 0111.1000       |

### Common mistake

Students may write 111.1 for 7.5. The task asks for exactly 4 integer bits and 4 fractional bits, so it should be 0111.1000.

## 9. Limits of Binary Representation

| Question                                      | Expected answer | Why                                   |
|-----------------------------------------------|-----------------|---------------------------------------|
| Largest 8-bit unsigned number                 | 255             | Unsigned range is $0$ to $2^8 - 1$ .  |
| Largest 8-bit signed two's complement number  | 127             | Signed range is $-2^7$ to $2^7 - 1$ . |
| Smallest 8-bit signed two's complement number | -128            | The negative side includes -128.      |

Unsigned n-bit range:  $0$  to  $2^n - 1$

Signed two's complement n-bit range:  $-2^{(n-1)}$  to  $2^{(n-1)} - 1$

## 10. Truth Tables and Boolean Expressions

| Symbol                 | Meaning                        |
|------------------------|--------------------------------|
| $A \times B$           | A AND B                        |
| $A + B$                | A OR B                         |
| NOT A / $\overline{A}$ | Flips A                        |
| $A \text{ XOR } B$     | True when inputs are different |

| A | B | A AND B | A OR B | A XOR B |
|---|---|---------|--------|---------|
| 0 | 0 | 0       | 0      | 0       |
| 0 | 1 | 0       | 1      | 1       |
| 1 | 0 | 0       | 1      | 1       |
| 1 | 1 | 1       | 1      | 0       |

### OR vs XOR

OR means at least one input is 1. XOR means exactly one input is 1. So  $1 \text{ OR } 1 = 1$ , but  $1 \text{ XOR } 1 = 0$ .

## Compound expression: (A + B) x NOT C

Teach students to do it in steps: calculate A + B, calculate NOT C, then AND those results.

| A | B | C | A + B | NOT C | Output |
|---|---|---|-------|-------|--------|
| 0 | 0 | 0 | 0     | 1     | 0      |
| 0 | 0 | 1 | 0     | 0     | 0      |
| 0 | 1 | 0 | 1     | 1     | 1      |
| 0 | 1 | 1 | 1     | 0     | 0      |
| 1 | 0 | 0 | 1     | 1     | 1      |
| 1 | 0 | 1 | 1     | 0     | 0      |
| 1 | 1 | 0 | 1     | 1     | 1      |
| 1 | 1 | 1 | 1     | 0     | 0      |

Expected expression matching: Table A = XOR, Table B = AND, Table C = OR, Table D = NOT.

## 11. Raspberry Pi CLI Activities

| Activity                | Commands                              | What students should learn                                             |
|-------------------------|---------------------------------------|------------------------------------------------------------------------|
| Finding your way around | pwd, ls, ls -a, cd, cd .., cd ~       | The terminal has a current location and they can move through folders. |
| Managing files          | mkdir, touch, mv, cp, rm              | Create, rename, copy, and delete files/folders.                        |
| nano editor             | nano, cat, CTRL+O, Enter, CTRL+X      | Edit text from the terminal and verify file contents.                  |
| System info             | whoami, df -h, free -h                | Check username, storage, and RAM.                                      |
| Network info            | hostname -I, ping, ip addr / ifconfig | Find IP address and test connectivity.                                 |

### High-frequency support point

Students will get lost in folders. Ask them to run pwd and ls first. Then move one directory at a time.

## 12. CLI Activity Details and Fixes

### Navigation

```
pwd
ls
ls -a
cd folder-name
cd ..
cd ~
```

- If they see "No such file or directory", check spelling, capitalization, spaces, and current folder.
- The repo path is likely week02/03\_CLI, not necessarily "Week 02/CLI".

## File management

```
mkdir practice
cd practice
touch file1.txt file2.txt
mv file1.txt notes.txt
cp notes.txt notes_backup.txt
rm file2.txt
```

- Use ls before rm so students can see what they are deleting.
- If a file is missing, check pwd and ls before assuming the command failed.

## nano editor

```
nano notes.txt
# Write text
CTRL + O
ENTER
CTRL + X
cat notes.txt
```

### Nano rescue

Use Control, not Command. CTRL+O saves, Enter confirms, CTRL+X exits.

## System and network info

```
whoami
df -h
free -h
hostname -I
ping google.com
# Stop ping with CTRL + C
ip addr
```

### ifconfig issue

If ifconfig is not installed, use ip addr. ifconfig is older; ip addr is the modern replacement.

## 13. Git Integration for Week 2

```
git status
git add .
git commit -m "Completed week 2 activities"
git push
```

| Problem                     | Likely reason                                                | Fix                                                   |
|-----------------------------|--------------------------------------------------------------|-------------------------------------------------------|
| nothing to commit           | Files not saved, created outside repo, or already committed. | Run pwd, ls, git status. Check files are inside repo. |
| fatal: not a git repository | Student is in the wrong folder.                              | Run pwd and ls, then cd into the repo folder.         |
| Work not visible on GitHub  | They committed but did not push.                             | Run git push and check GitHub website.                |

## 14. Physical and Software Setup to Check

- Raspberry Pis are available and have working microSD cards.
- Power supplies, monitors, keyboards, and mice are ready if not using SSH.
- Wi-Fi or Ethernet works; test with `hostname -I` and `ping google.com`.
- Students can log into the Pi and open terminal.
- Git is installed; test with `git --version`.
- Students can access GitHub Classroom and know where their repo folder is.
- Screenshot/screencast requirements are clear.
- Confirm whether images/results folders need to be created by students.

## 15. Simple Beginner Explanations

| Concept     | Simple explanation                                                          |
|-------------|-----------------------------------------------------------------------------|
| Binary      | How computers represent information using only 0 and 1.                     |
| Bit         | One binary digit: either 0 or 1.                                            |
| Byte        | 8 bits.                                                                     |
| Hexadecimal | Base 16; a shorter way to write binary numbers.                             |
| Truth table | Shows every possible input and the output for a logic expression.           |
| Terminal    | Lets you control the computer by typing commands instead of clicking icons. |
| Directory   | Another word for a folder.                                                  |
| IP address  | The device's address on a network.                                          |

## 16. Common Student Mistakes and Fixes

| Mistake                                             | How to fix/explain                                                                                       |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Thinking binary 1011 means one thousand and eleven. | Explain: $1011 = 8 + 2 + 1 = 11$ decimal.                                                                |
| Writing $1 + 1 = 2$ in binary.                      | Explain: in binary, 2 is written as 10, so write 0 and carry 1.                                          |
| Mixing up OR and XOR.                               | OR allows 1 and 1. XOR means exactly one input is 1.                                                     |
| Not using enough bits in fixed-point.               | Respect the required number of integer and fractional bits.                                              |
| Getting stuck in nano.                              | CTRL+O, Enter, CTRL+X.                                                                                   |
| No such file or directory.                          | Run <code>pwd</code> and <code>ls</code> , then check path/spelling.                                     |
| Ping keeps running.                                 | Press CTRL+C.                                                                                            |
| ifconfig not found.                                 | Use <code>ip addr</code> .                                                                               |
| Screenshots not pushed.                             | Run <code>git status</code> , <code>git add .</code> , <code>git commit</code> , <code>git push</code> . |

## 17. What You Should Practise Before the Lab

### Practise number systems

- Decimal to binary.



- Binary to decimal.
- Binary to octal.
- Binary to hexadecimal.
- Octal to decimal.
- Hexadecimal to decimal.

## Practise CLI on Raspberry Pi

```
pwd
ls
ls -a
cd ~
mkdir practice
cd practice
touch file1.txt file2.txt
mv file1.txt notes.txt
cp notes.txt notes_backup.txt
rm file2.txt
nano notes.txt
cat notes.txt
whoami
df -h
free -h
hostname -I
ping google.com
ip addr
```

## Practise Git workflow

```
git status
git add .
git commit -m "Completed week 2"
git push
```

## 18. Missing or Unclear Parts in the Repo

| Issue                                           | Why it matters                                              | Clarify with staff                                    |
|-------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|
| student/answers.json is mentioned but missing.  | Students may not know where to write number system answers. | Should they create it, or answer in Markdown?         |
| CLI path examples may not match repo structure. | Students may cd into folders that do not exist.             | Should they follow week02/03_CLI or the example path? |
| Screenshot/screencast wording is inconsistent.  | A screencast is video, but repo mentions .jpg images.       | Are screenshots or videos required?                   |
| ifconfig may not be installed.                  | Students may think their Pi is broken.                      | Is ip addr acceptable?                                |

## 19. Short Checklist Before Week 2 Lab

- Explain binary, decimal, octal, and hexadecimal.
- Convert between bases and check expected answers quickly.
- Explain binary carry and borrow.
- Explain two's complement as flip bits + add 1.
- Explain fixed-point binary and required bit lengths.

- Explain 8-bit unsigned and signed ranges.
- Complete AND, OR, NOT, XOR truth tables.
- Explain OR vs XOR clearly.
- Use Raspberry Pi terminal commands confidently.
- Help students navigate folders with `pwd`, `ls`, and `cd`.
- Help students create/edit/delete files.
- Help students use nano and stop ping.
- Help if `ifconfig` is missing.
- Help students commit and push work to GitHub.

**Main thing to remember**

For Week 2, your main job is helping students with binary/logic fundamentals and Raspberry Pi terminal confidence. Debug one step at a time.